Раздел 1

1. Подключитесь к серверу для выполнения практических работ, используя клиент для протокола ssh: ssh, putty, smartty (попробуйте их все и остановитесь на наиболее удобном для вас).

```
lksih@MacBook-Air-Lksih ~ % ssh -p 44556 eltex-pg1-v11@217.71.138.1
eltex-pg1-v11@217.71.138.1's password:
Linux eltex-2025-autumn 6.1.0-40-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.153-1
(2025-09-20) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Thu Mar  5 17:20:20 2026 from 217.71.131.234
```

2. Запустите новую сессию утилиты screen, с именем по умолчанию, посмотрите список подключенных пользователей командой w, отключитесь от сессии.

```
eltex-pg1-v11@eltex-2025-autumn:~$ screen

eltex-pg1-v11@eltex-2026-spring:~$ w
 17:58:09 up  6:35, 56 users,  load average: 0.04, 0.05, 0.10
USER     TTY      FROM             LOGIN@   IDLE   JCPU   PCPU WHAT
root     tty1     -                11:23    6:34m  0.02s   ?    -bash
tech     pts/0    217.71.139.234   17:14    8:50   0.05s  0.02s sshd: tech [pri
tech     pts/1    37.193.179.176   11:36    6:21m  0.01s  0.02s sshd: tech [pri
eltex-pg pts/2    217.71.131.234   17:24   25.00s  0.04s  0.04s -bash
eltex-pg pts/3    217.71.131.234   17:30   57.00s  0.01s   ?    screen -r top
eltex-pg pts/5    217.71.131.234   17:26    1.00s  0.00s   ?    screen
eltex-pg pts/6    :pts/8:S.2       17:51   14:11   0.00s   ?    /usr/bin/bash
eltex-pg pts/7    217.71.139.234   17:23    1.00s  0.04s   ?    screen -S test3
eltex-pg pts/8    217.71.131.234   17:45    1:29   0.01s   ?    screen -r 36025
eltex-pg pts/9    217.71.131.234   17:24   19.00s  0.01s   ?    screen
eltex-pg pts/10   91.78.233.217    17:25    4.00s  0.02s   ?    screen -r 36258
eltex-pg pts/13   :pts/20:S.2      17:35   20:55   0.00s   ?    /usr/bin/bash
eltex-pg pts/15   :pts/50:S.0      17:48    3:52   0.94s  0.17s SCREEN -dmS top
eltex-pg pts/16   217.71.131.234   17:46    1.00s  0.01s   ?    screen
eltex-pg pts/20   89.46.235.225    17:31    1.00s  0.01s   ?    screen
eltex-pg pts/21   :pts/20:S.0      17:33   24:09   0.00s  0.19s SCREEN
eltex-pg pts/18   217.71.131.234   17:33    1.00s  0.01s   ?    screen -r 36363
eltex-pg pts/24   217.71.131.234   17:33    1.00s  0.00s   ?    screen
eltex-pg pts/25   :pts/20:S.1      17:34    1.00s  0.00s   ?    /usr/bin/bash
eltex-pg pts/26   217.71.139.234   17:39   16:49   0.00s   ?    -bash
eltex-pg pts/29   :pts/24:S.0      17:36    5:45   0.01s   ?    screen -d -r to
eltex-pg pts/30   :pts/24:S.1      17:37    1.00s  0.00s   ?    /usr/bin/bash
eltex-pg pts/32   :pts/20:S.3      17:37    8:57   0.20s  0.20s watch /usr/bin/
```

```
eltex-pg pts/33   :pts/29:S.0      17:52   16:05   1.17s  0.12s SCREEN -dmS top
eltex-pg pts/34   :pts/17:S.0      17:37   14:17   0.01s   ?    screen -S top
eltex-pg pts/36   :pts/18:S.0      17:57    1.00s  0.00s  1.94s SCREEN -S -d -m
eltex-pg pts/38   :pts/16:S.0      17:48    1.00s  0.00s   ?    screen -S top
eltex-pg pts/39   :pts/43:S.0      17:57    1:56   0.00s  0.03s SCREEN -S top
eltex-pg pts/41   :pts/34:S.0      17:41   14:17   0.00s  0.02s SCREEN -S top
eltex-pg pts/35   :pts/10:S.0      17:53    1:53   0.01s   ?    SCREEN -S top -
eltex-pg pts/43   91.79.228.242    17:42   45.00s  0.01s   ?    screen -r top
eltex-pg pts/44   :pts/8:S.0       17:51   14:25   0.00s   ?    SCREEN
eltex-pg pts/45   :pts/3:S.1       17:54    3:05   0.08s  0.07s watch /usr/bin/
eltex-pg pts/46   :pts/8:S.1       17:51    1:29   0.00s   ?    /usr/bin/bash
eltex-pg pts/47   188.162.13.15    17:45   57.00s  0.03s   ?    screen -r whoam
eltex-pg pts/48   217.71.139.234   17:44   11:19   0.01s  0.01s -bash
eltex-pg pts/50   :pts/20:S.4      17:45    3:52   0.01s   ?    screen -r top
eltex-pg pts/55   :pts/60:S.0      17:54    1.00s  0.00s   ?    SCREEN -S top
eltex-pg pts/60   :pts/38:S.0      17:50    1.00s  0.00s   ?    screen -S top
eltex-pg pts/61   217.71.131.234   17:50    6:04   0.01s  0.01s -bash
eltex-pg pts/62   :pts/5:S.0       17:57    1.00s  0.19s  0.19s w
eltex-pg pts/63   :pts/3:S.0       17:54    3:45   0.56s  0.09s SCREEN -S top -
eltex-pg pts/53   91.78.233.73     17:52    1:11   0.01s   ?    screen
eltex-pg pts/66   :pts/49:S.0      17:53    2:25   0.00s   ?    SCREEN -S test3
eltex-pg pts/68   178.130.142.199  17:53    1:29   0.00s   ?    -bash
eltex-pg pts/70   :pts/20:S.5      17:54    2:33   0.06s  0.06s watch /usr/bin/
eltex-pg pts/71   :pts/47:S.0      17:54   57.00s  0.01s   ?    SCREEN -S whoam
eltex-pg pts/72   :pts/7:S.0       17:57    1.00s  0.00s   ?    SCREEN -S test3
eltex-pg pts/73   217.71.131.234   17:55    8.00s  0.00s   ?    -bash
eltex-pg pts/74   :pts/49:S.1      17:55    2:09   0.00s   ?    /usr/bin/bash
eltex-pg pts/79   217.71.131.234   17:56    1:39   0.00s   ?    -bash
eltex-pg pts/80   :pts/10:S.1      17:56    4.00s  0.00s   ?    /usr/bin/bash
eltex-pg pts/81   217.71.139.234   17:56    1:29   0.00s   ?    -bash
eltex-pg pts/82   :pts/53:S.0      17:56    1:16   0.00s   ?    SCREEN
eltex-pg pts/83   :pts/9:S.0       17:56   19.00s  0.00s   ?    vi report_part1
eltex-pg pts/85   :pts/43:S.1      17:57   45.00s  0.00s   ?    watch /usr/bin/

[screen is terminating]
```

3. Запустите отсоединенную сессию утилиты screen, при этом запустите в этой сессии команду top. Назовите сессию именем «top».

```
screen -S top -d -m top
```

4. Получите список сессий, созданных утилитой screen.

```
screen -ls
There is a screen on:
        45706.top  (03/05/2026 06:01:52 PM)    (Detached)
1 Socket in /run/screen/S-eltex-pg1-v11.
```

5. Подсоединитесь к сессии top.

screen -r top

```
top - 18:04:59 up  6:42, 68 users,  load average: 0.02, 0.02, 0.07
Tasks: 405 total,   1 running, 404 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.4 us,  0.5 sy,  0.0 ni, 98.9 id,  0.0 wa,  0.0 hi,  0.2 si,  0.0 st
MiB Mem :   3914.7 total,   2294.1 free,    793.7 used,   1066.4 buff/cache
MiB Swap:    976.0 total,    976.0 free,      0.0 used.   3121.0 avail Mem


  35246 eltex-p+  20   0   10740   5132   1968 S   0.3   0.1   0:00.33 screen
  35327 eltex-p+  20   0   11640   5548   3416 S   0.3   0.1   0:02.64 top
  37505 eltex-p+  20   0   11640   5500   3356 S   0.3   0.1   0:01.70 top
  37537 eltex-p+  20   0   11640   5464   3312 S   0.3   0.1   0:01.53 top
  45351 eltex-p+  20   0   11640   5560   3416 S   0.3   0.1   0:00.31 top
  47382 eltex-p+  20   0    8268   2620   1904 S   0.3   0.1   0:00.02 screen
      1 root      20   0  168644  13044   9204 S   0.0   0.3   0:01.43 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kthreadd
      3 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 rcu_gp
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 rcu_par_gp
      5 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 slub_flushwq
      6 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 netns
     10 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 mm_percpu_wq
     11 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_kthread
     12 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_rude_kthread
     13 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_trace_kthread
     14 root      20   0       0      0      0 S   0.0   0.0   0:00.12 ksoftirqd/0
     15 root      20   0       0      0      0 I   0.0   0.0   0:00.48 rcu_preempt
     16 root      rt   0       0      0      0 S   0.0   0.0   0:00.07 migration/0
     18 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/0
     19 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/1
     20 root      rt   0       0      0      0 S   0.0   0.0   0:00.38 migration/1
     21 root      20   0       0      0      0 S   0.0   0.0   0:00.09 ksoftirqd/1
     23 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/1:0H-events_highpri
     24 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/2
     25 root      rt   0       0      0      0 S   0.0   0.0   0:00.38 migration/2
     26 root      20   0       0      0      0 S   0.0   0.0   0:00.08 ksoftirqd/2
     28 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/2:0H-events_highpri
     29 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/3
     30 root      rt   0       0      0      0 S   0.0   0.0   0:00.38 migration/3
     31 root      20   0       0      0      0 S   0.0   0.0   0:00.09 ksoftirqd/3
     33 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/3:0H-events_highpri
     38 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kdevtmpfs
     39 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 inet_frag_wq
     40 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kauditd
     41 root      20   0       0      0      0 S   0.0   0.0   0:00.00 khungtaskd
     42 root      20   0       0      0      0 S   0.0   0.0   0:00.00 oom_reaper
     43 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 writeback
     44 root      20   0       0      0      0 S   0.0   0.0   0:00.44 kcompactd0
```

```
 45 root      25   5       0      0      0 S   0.0   0.0   0:00.00 ksmd
 46 root      39  19       0      0      0 S   0.0   0.0   0:00.17 khugepaged
 47 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kintegrityd
 48 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kblockd
 49 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 blkcg_punt_bio
 50 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 tpm_dev_wq
 51 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 edac-poller
 52 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 devfreq_wq
 54 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/0:1H-kblockd
 55 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kswapd0
 62 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kthrotld
 64 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 acpi_thermal_pm
 66 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 mld
 67 root       0 -20       0      0      0 I   0.0   0.0   0:00.02 kworker/2:1H-kblockd
 68 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 ipv6_addrconf
 73 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kstrp
 78 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 zswap-shrink
 79 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/u9:0
124 root       0 -20       0      0      0 I   0.0   0.0   0:00.05 kworker/1:1H-kblockd
134 root       0 -20       0      0      0 I   0.0   0.0   0:00.05 kworker/3:1H-kblockd
153 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 ata_sff
154 root      20   0       0      0      0 S   0.0   0.0   0:00.00 scsi_eh_0
```

6. Создайте еще одно окно в сессии top.

```
eltex-pg1-v11@eltex-2026-spring:~$
```

7. В созданном окне выполните команду:
8. watch /usr/bin/vmstat

```
Every 2.0s: /usr/bin/vmstat          eltex-2026-spring: Thu Mar  5 18:07:41 2026
```

```
procs -----------memory---------- ---swap-- -----io---- -system-- ------cpu-----
 r  b   swpd   free   buff  cache   si   so    bi    bo   in   cs us sy id wa st
 1  0      0 2344420  69996 1022528    0    0     5    19   17   22  0  0 99  1
 0
```

9. Получите список окон.

```
Num Name                                                    Flags

  0 top                                                    $
  1 bash                                                     $
```

10. Переименуйте второе окно как «vmstat».

```
Num Name                                                    Flags

  0 top                                                    $
  1 vmstat                                                   $
```

11. Перейдите в окно, в котором работает top.

```
top - 18:30:46 up  7:08, 59 users,  load average: 0.03, 0.06, 0.09
Tasks: 384 total,   1 running, 381 sleeping,   2 stopped,   0 zombie
%Cpu(s):  0.3 us,  0.8 sy,  0.0 ni, 95.8 id,  2.9 wa,  0.0 hi,  0.2 si,  0.0 st
MiB Mem :   3914.7 total,   2333.4 free,    749.4 used,   1073.2 buff/cache
MiB Swap:    976.0 total,    976.0 free,      0.0 used.   3165.4 avail Mem


  35327 eltex-p+  20   0   11640   5548   3416 S   0.3   0.1   0:05.64 top
  38887 eltex-p+  20   0   11640   5568   3420 S   0.3   0.1   0:03.49 top
  42906 eltex-p+  20   0    6724   3416   2444 S   0.3   0.1   0:00.76 watch
  53191 eltex-p+  20   0   11632   5564   3416 S   0.3   0.1   0:03.67 top
  55649 eltex-p+  20   0   18104   7100   5200 S   0.3   0.2   0:00.16 sshd
  75787 eltex-p+  20   0    6584   3332   2472 S   0.3   0.1   0:00.13 watch
  79876 eltex-p+  20   0    6184   2892   2472 S   0.3   0.1   0:00.03 watch
  83306 eltex-p+  20   0   11640   5532   3388 S   0.3   0.1   0:00.06 top
      1 root      20   0  168644  13048   9204 S   0.0   0.3   0:01.57 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kthreadd
      3 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 rcu_gp
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 rcu_par+
      5 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 slub_fl+
      6 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 netns
     10 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 mm_perc+
```

12. Отсоединитесь от сессии top.

```
[detached from 45706.top]
```

13. Запустите новую сессию и запустите в окне команду:

vi report_part1.txt

screen -S vi

14. Отсоединитесь от сессии и получите список сессий.

```
screen -ls
There are screens on:
        91701.vi     (03/05/2026 06:36:32 PM)    (Detached)
        45706.top   (03/05/2026 06:01:52 PM)    (Detached)
2 Sockets in /run/screen/S-eltex-pg1-v11.
```

15. Отключитесь от сервера, используя команду logout или комбинацию клавиш Ctrl+D

```
logout
Connection to 217.71.138.1 closed.
```

16. Подключитесь к серверу (как в п.1), и получите список сессий screen

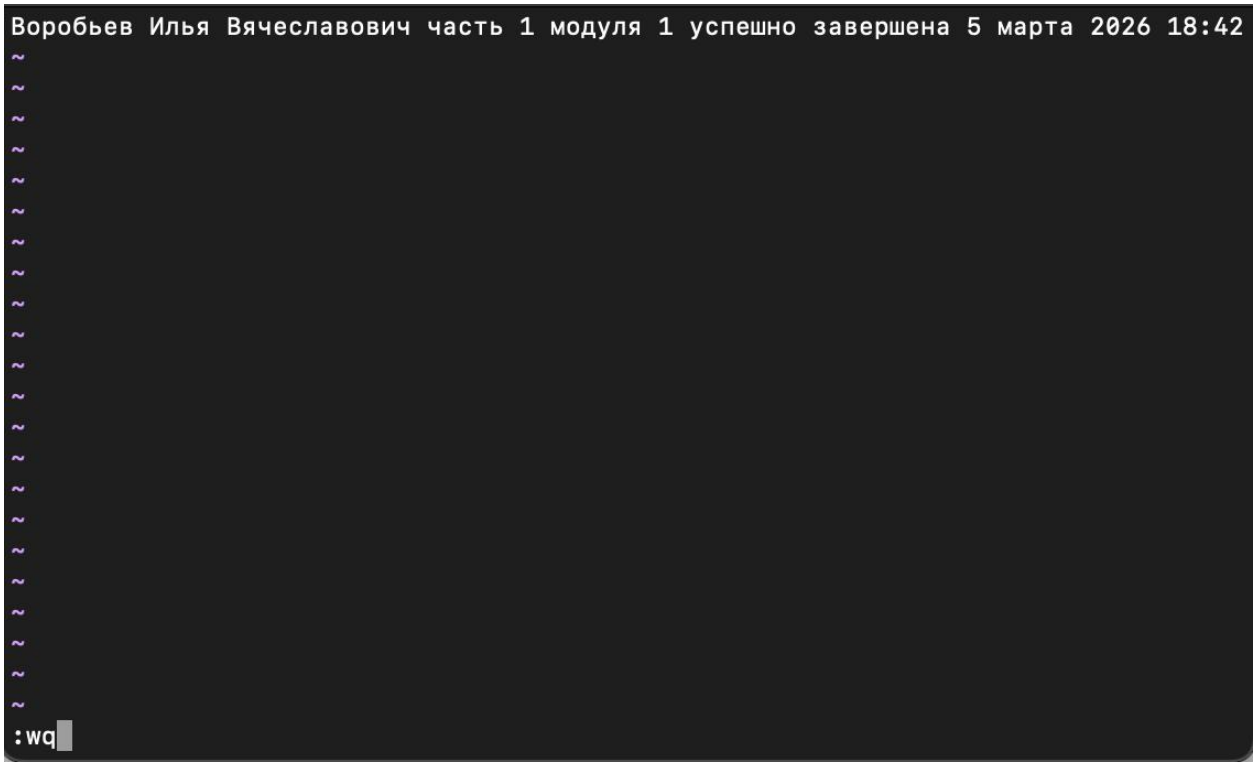
```
ssh -p 44556 eltex-pg1-v11@217.71.138.1
eltex-pg1-v11@217.71.138.1's password:
Linux eltex-2026-spring 6.1.0-40-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.153-1
(2025-09-20) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.
```

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent permitted by applicable law.
Last login: Thu Mar 5 17:26:19 2026 from 217.71.131.234

```
screen -ls
There are screens on:
        91701.vi     (03/05/2026 06:36:33 PM)    (Detached)
        45706.top  (03/05/2026 06:01:53 PM)    (Detached)
2 Sockets in /run/screen/S-eltex-pg1-v11.
```

17. Подключитесь к сессии с редактором и запишите в файл "ФИО часть 1 модуля 1 успешно завершена дата и время"

```
screen -r vi
```



18. Закройте редактор, сохранив файл и закройте все сессии screen.

```
"report_part1.txt" [New] 1L, 137B written

[detached from 91701.vi]

screen -X -S vi quit
screen -X -S top quit
screen -ls
No Sockets found in /run/screen/S-eltex-pg1-v11.
```